

SUMMARY

My research focuses on **continual learning and adaptation** in language models and **AI agents**. I investigate how models **learn from imperfect supervision**, **personalize over long horizons**, and **support trustworthy evaluation**. My goal is to develop AI systems that continuously improve after deployment while retaining acquired knowledge and learning to correct their own mistakes.

EDUCATION

Yale University

Master of Science, Computer Science | Co-advisors: Dr. Arman Cohan, Dr. Rex Ying
• Full Scholarship + Teaching Fellowship

New Haven, CT
Aug 2025 – May 2027

Vanderbilt University

Bachelor of Science, Computer Science & Mathematics | Minor: Data Science
• GPA: 3.99/4.00; Honors: *summa cum laude* (Top 5%), *Dean's List* all semesters

Nashville, TN
Aug 2022 – May 2025

University of California, Irvine

Bachelor of Science, Computer Science
• GPA: 4.00/4.00; Honors: *Dean's List* all quarters

Irvine, CA
Sep 2021 – Jun 2022

PATENTS AND PUBLICATIONS

C=Conference, W=Workshop, J=Journal, S=In Submission

- [S.1] Leyao Wang, Rex Ying, Arman Cohan. **Stronger, Yet Not Better: Demystifying Self-Distillation Initialization**. (in submission to *ICLR* 2027)
- [S.2] Leyao Wang*, Yanan He*, Peng Chen, Asaf Yehudai, Yixin Liu, Rex Ying, Michal Shmueli-Scheuer, Arman Cohan. **Time to REFLECT: Can We Trust LLM Judges for Evidence-based Research Agents**. (in submission to *ARR* 2026)
- [S.3] Xutao Mao, Liangjie Zhao, Leyao Wang, Rui Qian, Qiang Huang, Wentao Wang, Bo Han, Xiang Zheng, Cong Wang. **Agents Don't Just Agree, They Remember: Benchmarking Persistent Sycophancy in Stateful Personal Agents**. (in submission to *ARR* 2026)
- [S.4] Leyao Wang*, Xutao Mao*, Xuhui Zhan, Yuying Zhao, Bo Ni, Ryan A. Rossi, Nesreen K. Ahmed, Tyler Derr. **Towards Bridging Review Sparsity in Recommendation with Textual Edge Graph Representation**. (in submission to *WSDM* 2027)
- [C.1] Yangtian Zhang*, Leyao Wang*, Hiren Madhu, Ngoc Bui, Walter Roznyatovskiy, Rex Ying. **CoPersona: Collaborative Persona Graphs for Robust LLM Personalization**. *KDD* 2026.
- [C.2] Leyao Wang, Yu Wang, Bo Ni, Yuying Zhao, Hanyu Wang, Yao Ma, Tyler Derr. **SaVe-TAG: LLM-based Interpolation for Long-Tailed Text-Attributed Graphs**. *KDD* 2026.
- [C.3] Xutao Mao, Ezra Xuanru Tao, Leyao Wang. **MindVote: When AI Meets the Wild West of Social Media Opinion**. *AAAI* 2026 (**Oral**).
- [C.4] Bo Ni, Leyao Wang, ..., Tyler Derr, Ryan A Rossi. **Large Language Models for Conversational User Simulation: A Comprehensive Survey**. *EACL* 2026.
- [J.1] Leyao Wang*, Zhiyu Wan*, Congning Ni, Qingyuan Song, Yang Li, Ellen W. Clayton, Bradley Malin, Zhijun Yin. **Application and Concerns of ChatGPT and Conversational Large Language Models in Healthcare: A Systematic Review**. *JMIR*.
- [J.2] Wenqiang Chen*, Jason Cheng*, Leyao Wang, Wei Zhao, Wojciech Matusik (2024). **Sensor2Text: Enabling Natural Language Interactions for Daily Activity Tracking Using Wearable Sensors**. *IMWUT*.
- [J.3] Bo Ni, Zheyuan Liu[†], Yongjia Lei[†], Leyao Wang[†], ... Yu Wang*, Meng Jiang*, Tyler Derr*. **Towards Trustworthy Retrieval Augmented Generation for Large Language Models: A Survey**. *CSUR*.
- [W.1] Leyao Wang*, Rishab Pulugurta*, Pranay Vure*, Yinuo Zhang*, Aastha Pal, Pranam Chatterjee (2024). **PepDoRA: A Unified Peptide Language Model via Weight-Decomposed Low-Rank Adaptation**. *AIDrugX @ NeurIPS* 2024.

Notes: * equal contribution; [†] alphabetical co-first/co-equal as noted.

RESEARCH & INDUSTRY EXPERIENCE

YaleNLP Lab, Yale University

Graduate Researcher | Co-advisor: Dr. Arman Cohan

New Haven, CT
May 2025 – Present

- Investigated the reliability of **LLM-as-a-judge** for evidence-based **deep research agents**, examining whether automated evaluations faithfully assess evidence quality, factual grounding, and research outcomes.
- Studied **SFT** warm-up and initialization strategies for on-policy self-distillation (**OPSD**), analyzing how initial policy capabilities shape subsequent learning trajectories and post-training performance.
- Examined the relationship between model initialization, self-generated supervision, and training dynamics to understand when stronger initial models lead to effective continued improvement.

Graph and Geometric Learning Lab, Yale University

Graduate Researcher | Co-advisor: Dr. Rex Ying

New Haven, CT
May 2025 – Present

- Developed collaborative persona graphs for robust **LLM personalization**, representing user characteristics and relationships through structured knowledge to support adaptation under sparse observations.
- Leveraged collaborative user representations for **soft prompt tuning**, enabling LLMs to generate personalized responses by capturing shared information across related users and personas.
- Studied persistent **sycophancy** in stateful **personal agents**, examining how inaccurate conversational information can enter long-term memory and influence behavior across future interactions.

Network and Data Science Lab, Vanderbilt University

Undergraduate Researcher | Advisor: Dr. Tyler Derr

Nashville, TN
Jan 2024 – Aug 2025

- Developed LLM-based interpolation for **long-tailed** text-attributed graphs, generating informative representations to enrich **sparse supervision** and improve learning on underrepresented nodes.
- Investigated sparse user-item interactions through textual edge representations, exploring how language-based information can support **personalized recommendation** under limited observations.
- Studied **trustworthy** and **human-centered** LLM systems through conversational user simulation and retrieval-augmented generation (**RAG**), examining user modeling and reliability challenges.

Chatterjee Lab, Duke University

Undergraduate Researcher | Advisor: Dr. Pranam Chatterjee

Remote
May 2024 – Dec 2024

- Developed parameter-efficient adaptation of peptide language models using weight-decomposed low-rank methods, a variant of **LoRA**, for unified representation and generation.

Institute for Software Integrated Systems (ISIS), Vanderbilt University

Research Intern | Advisor: Prof. Gautam Biswas

Nashville, TN
Apr 2024 – Jun 2024

- Developed a **multi-modal** pipeline using WhisperX and GPT-4o to extract behavioral, sentiment, and affective signals from spoken interactions.

Computer Science and Artificial Intelligence Laboratory, MIT

Research Assistant | Mentor: Dr. Wenqiang Chen

Remote
Mar 2024 – May 2024

- Developed **transfer learning** methods to translate heterogeneous wearable-sensor observations into natural-language representations of daily activities.
- Applied **SFT** to develop language-based interfaces for personalized interpretation and interaction with **multi-modal** activity-tracking data.

SPHERE Lab, Vanderbilt University

Undergraduate Researcher | Advisor: Dr. Zhijun Yin

Nashville, TN
Apr 2023 – Apr 2024

- Conducted a systematic review of conversational LLM applications in clinical and consumer healthcare, evaluating reliability, privacy, and ethical challenges to **trustworthy** deployment.

Metissa.AI

Research & Development Intern

Remote
Apr 2023 – Aug 2023

- Developed a GPT-3.5-powered educational chatbot using retrieval-augmented generation (**RAG**) and model customization to improve response relevance and accuracy.

HONORS AND AWARDS

Dean's Award for Outstanding Scholarship (summa cum laude, Top 5% in Vanderbilt University)

May 2025

Honorable Mention, CRA Outstanding Undergraduate Award

Jan 2025

Vanderbilt University Summer Research Program (Stipend: \$6000)

Apr 2024

Data Science Institute Research Program (Stipend: \$6000)

Apr 2024

ISIS Summer Internship Program (Stipend: \$8000)

Apr 2024

PROFESSIONAL SERVICE

Reviewer: AAAI '26, KDD '25; ACL '26; TKDD; TAI; JMIR; MLoG-GenAI@KDD '25

Teaching Assistant: Trustworthy DL (Yale); Intro to LLM (Yale); Stats and Prob (Vanderbilt)